



Sydney Technical High School

HSC Assessment Task 1

December 2011

Mathematics

Time allowed – 70 minutes

Instructions

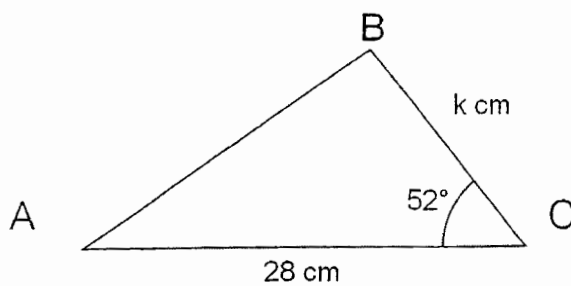
- Use a blue or black pen.
- Approved calculators may be used.
- All necessary working must be shown. Marks may not be awarded for careless or badly arranged work.
- Total marks – 55
- Attempt all questions, using the booklets provided.
- Marks awarded are shown on each question.
- Start each question on a new page.

Question 1

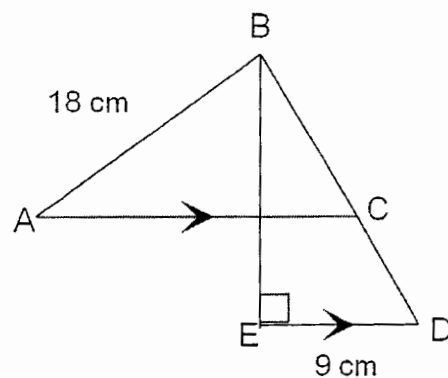
8 Marks

- a) Solve $\cos x = -\frac{1}{\sqrt{2}}$ for $0^\circ \leq x \leq 360^\circ$ 2
- b) Given $\tan \theta = \frac{-5}{12}$, find $\sin \theta$, if θ is obtuse. 1
- c) The area of the triangle is 253.74 cm^2 and $\angle ACB = 52^\circ$. 2

Calculate the value of k to the nearest centimetre.



- d) In the diagram, $AB = 16 \text{ cm}$, $AC = 20 \text{ cm}$, $BE = 15 \text{ cm}$ and $ED = 9 \text{ cm}$.
 $BE \perp ED$ and $AC \parallel ED$.
- (i) Find the size of $\angle BDE$, to the nearest degree. 1
- (ii) Using the Sine Rule, find the size of $\angle ABC$, to the nearest degree. 2



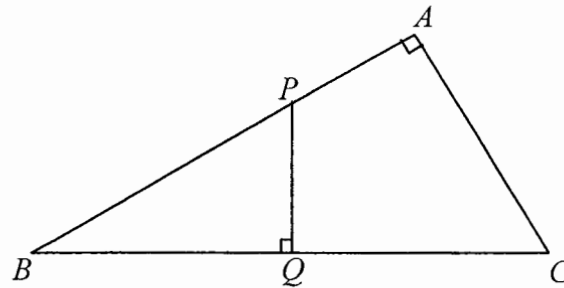
Question 2

8 Marks

- a) A point $P(x,y)$ moves so that the sum of the squares of its distance from each of the points $A(0,0)$ and $B(4,0)$ is equal to 40. 3

Show that the locus of $P(x,y)$ is a circle, and state its radius and centre.

b)



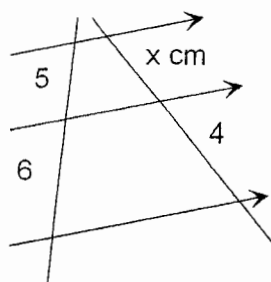
In the diagram, ABC is a right angled triangle with $AB = 8$ cm and $AC = 6$ cm.

If the triangle is folded along the line PQ , vertex B coincides with vertex C .

- (i) Show that triangles ABC and QBP are similar. 2

- (ii) Find the length of PQ . 1

- c) Find the value of x in the diagram. (Give a reason) 2



Question 3**8 Marks**

- a) The first three terms of a sequence are 7, 13 and 19.
- (i) Is this sequence arithmetic, geometric or neither? (Show reasoning) 1
 - (ii) Find a simplified expression for the n th term, T_n . 1
 - (iii) Find T_{32} . 1
 - (iv) Is 243 a term of the sequence? (Show appropriate working.) 2
- b) Find the values of A , B and C if $3x^2 + 5x \equiv A(x + 1)^2 + B(x + 1) + C$ 3

Question 4**7 Marks**

- a) (i) The fourth term of a geometric sequence is $\frac{2}{9}$ and the seventh term is $\frac{2}{243}$. 3
Find the first three terms.
- (ii) Explain why a limiting sum exists. (Show reasoning) 1
- (iii) Find the limiting sum (sum to infinity). 1
- b) Find the geometric series whose first term is 3 and whose limiting sum is 9. 2

Question 5**9 Marks**

- a) The equation of a parabola is given as $(x - 2)^2 = 4y$.
- (i) Find the coordinates of the focus. 1
 - (ii) What is the equation of the directrix? 1
 - (iii) Sketch the parabola, showing all important features. 2
- b) Find the equation of the parabola with vertex $(-1, 2)$, axis parallel to the x axis and passing through the point $(7, 10)$. 2
- c) Find the equation of the tangent to the parabola $x^2 = -16y$ at the point $(-2, -\frac{1}{4})$. 3

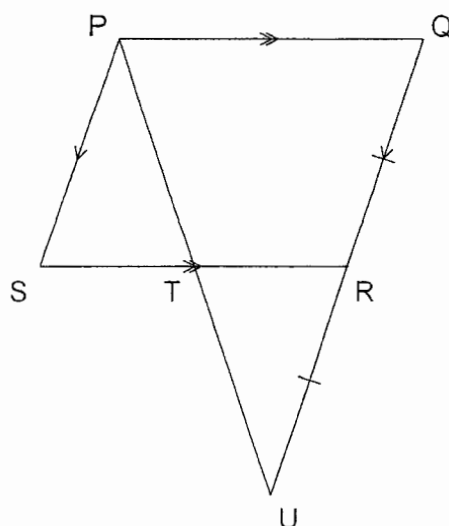
Question 6**8 Marks**

- a) For the parabola with equation $y = 10 + 3x - x^2$, find: 4
- (i) the equation of the axis of symmetry.
 - (ii) the coordinates of the vertex.
 - (iii) the x intercepts.
 - (iv) the values of x for which $10 + 3x - x^2 > 0$.
- b) Write a quadratic equation with roots α and β if $\alpha + \beta = -2$ and $\alpha\beta = 6$. 1
- c) Make a suitable substitution and solve $(x^2 - 2)^2 - 7(x^2 - 2) = 0$. 3

Question 7

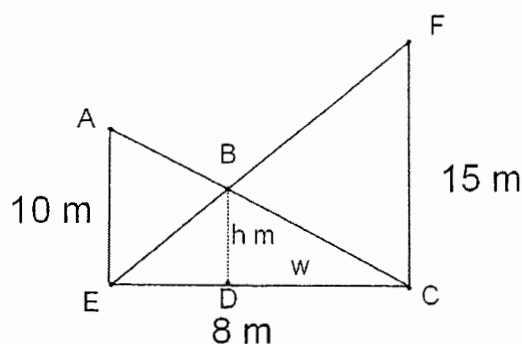
7 Marks

- a) In the diagram, $PQRS$ is a parallelogram. QR is produced to U so that $QR = RU$.



- (i) Giving appropriate reasons, prove that $\triangle PST$ and $\triangle URT$ are congruent. 3
 - (ii) Hence or otherwise, show that T is the midpoint of SR . 1
- b) Two vertical poles of height 10m and 15m are 8m apart. Wire stretches from the top of each pole to the foot of the other. See diagram below.

- (i) Use the properties of similar figures to find an expression for w in terms of h . [Hint: Let $DC = w$] 1
- (ii) Find the height, $h\text{ m}$ above the ground where the wires cross. 2



End of Paper

HSC ASSESSMENT 1 SOLUTIONS MATHEMATICS 2011

Q1 a) $\cos x = -\frac{1}{\sqrt{2}}$

$\therefore x = 135^\circ, 225^\circ$

b) $\tan \theta = -\frac{5}{12}$

$\therefore \sin \theta = \frac{5}{13}$

c) $A = \frac{1}{2} ab \sin C$

$\therefore 253.74 = \frac{1}{2} \times 28 \times k \times \sin 52^\circ$

$= 14k \sin 52^\circ$

$\therefore k = \frac{253.74}{14 \sin 52^\circ}$

$\therefore 23 \text{ cm}$

d) (i) $\tan \widehat{BDE} = \frac{15}{9\frac{2}{3}}$

$= 1\frac{2}{3}$

$\therefore \widehat{BDE} = 59^\circ$

(ii) $\therefore \widehat{ACB} = 59^\circ$ (corresponding $\angle$ s)

$\therefore \frac{\sin \widehat{ABC}}{20} = \frac{\sin 59^\circ}{18}$

$\therefore \sin \widehat{ABC} = \frac{10 \sin 59^\circ}{9}$

$= 72^\circ 15'$

$\therefore 72^\circ$

Q2 a) $(x-0)^2 + y^2 + (x-4)^2 + y^2 = 40$

$\therefore x^2 + y^2 + x^2 - 8x + 16 + y^2 = 40$

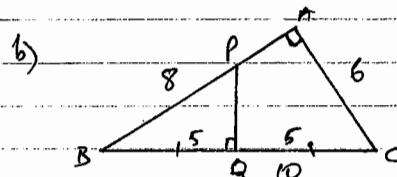
$\therefore 2x^2 + 2y^2 - 8x = 24$

$\therefore x^2 + y^2 - 4x = 12$

$\therefore (x-2)^2 + y^2 = 16$

circle: centre (2,0)

radius 4



(i) In Δ s QBP and ABC,

$\widehat{ABC}$ is common

$\widehat{BQP} = \widehat{BAC} = 90^\circ$

$\therefore \Delta QBP \parallel \Delta ABC$ (AAA)

(ii) $\frac{PQ}{BQ} = \frac{6}{8}$

$\therefore PQ = 5 \times \frac{6}{8}$

$= 3.75 \text{ cm}$

Intercepts in proportion across parallel lines.

Q3

For AP

a) (i) $13 - 7 = 6$

$19 - 13 = 6$

$\therefore$ AP

For GP

(ii) $T_n = a + (n-1)d$

$= 7 + 6(n-1)$

$= 6n + 1$

$\therefore$ Not GP

(iv) Let $T_n = 243$

$\therefore 6n + 1 = 243$

$\therefore 6n = 242$

$\therefore n = \frac{242}{6}$

$= 40\frac{1}{3}$

243 is not a term as $40\frac{1}{3} \notin \mathbb{N}$

b) $3x^2 + 5x = A(x+1)^2 + B(x+1) + C$

RHS = $A(x+1)^2 + B(x+1) + C$

$= A(x^2 + 2x + 1) + Bx + B + C$

$= Ax^2 + 2Ax + A + Bx + B + C$

$= Ax^2 + (2A+B)x + (A+B+C)$

$\therefore A = 3$

$2A+B=5$

$\therefore B = -1$

$A+B+C=0$

$\therefore 3-1+C=0$

$\therefore C = -2$

4b) $a=3$

$\frac{9}{1-r} = 9$

$\therefore \frac{3}{1-r} = 9$

$\therefore \frac{1}{1-r} = 3$

$\therefore 1-r = \frac{1}{3}$

$\therefore r = 1 - \frac{1}{3}$

$= \frac{2}{3}$

Seq $\Rightarrow 3, 2, \frac{4}{3}, \dots$

Q4 a) (i) $T_n = ar^{n-1}$

$\therefore T_4 = ar^3 = \frac{2}{9}$

$T_7 = ar^6 = \frac{2}{243}$

$\therefore \frac{ar^6}{ar^3} = \frac{\frac{2}{243}}{\frac{2}{9}} \times \frac{9}{2}$

$\therefore r^3 = \frac{1}{27}$

$\therefore r = \frac{1}{3}$

$\frac{9}{27} = \frac{2}{9}$

$\therefore a = \frac{2 \times 27}{9}$

$= 6$

Seq $\Rightarrow 6, 2, \frac{2}{3}$

(ii) Limiting sum exists

because $-1 < r < 1$ [or $|r| < 1$]

(iii) $S_\infty = \frac{a}{1-r}$

$= \frac{6}{1-\frac{1}{3}}$

$= 9$

Q5 a) $(x-2)^2 = 4y$

General form $\Rightarrow (x-x_1)^2 = 4a(y-y_1)$ (iii)

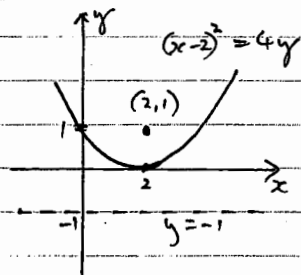
$\therefore a=1, x_1=2, y_1=0$

vertex $(2,0)$

(i) Parabola upright $\therefore$ focus $(2,1)$

(ii) Directrix $y=-1$

$\therefore y=-1$



b) $(y-y_1)^2 = 4a(x-x_1)$

$\therefore x_1=-1$

$y_1=2$

$(y-2)^2 = 4a(x+1)$

But $(7,10)$ satisfies

$\therefore (10-2)^2 = 4a(7+1)$

$64 = 4a \times 8$

$\therefore 4a = 8$

$\therefore a = 2$

$\therefore (y-2)^2 = 8(x+1)$

c) $x^2 = -16y$ $(-2, -\frac{1}{4})$

$y = -\frac{x^2}{16}$

$\therefore \frac{dy}{dx} = -\frac{2x}{16}$

$= -\frac{x}{8}$

$= \frac{1}{4}$ when $x=-2$

$\therefore$ equation of tangent $\Rightarrow$

$y-y_1 = m(x-x_1)$

$y+\frac{1}{4} = \frac{1}{4}(x+2)$

$\therefore 4y+1 = x+2$

$\therefore x-4y+1=0$

Q6 a) $y = -x^2 + 3x + 10$

(i) axis of symmetry when $x = -\frac{b}{2a}$

$\therefore x = \frac{-3}{2 \times -1}$

$\therefore x = \frac{3}{2}$

(ii) $x = \frac{3}{2}$

$\therefore y = -(\frac{3}{2})^2 + 3 \times \frac{3}{2} + 10$

$= -\frac{9}{4} + \frac{9}{2} + 10$

$= \frac{-9+18+40}{4}$

$\therefore$ vertex $\Rightarrow (\frac{3}{2}, 12\frac{1}{4})$

(iii) $-x^2 + 3x + 10 < 0$

$\therefore x^2 - 3x - 10 > 0$

$\therefore (x-5)(x+2) > 0$

$\therefore x < -2$ or $x > 5$

(iv) $a < 0$ $\therefore$ concave down

$\therefore$ positive when $-2 < x < 5$

Q6 b) $(x+\alpha)(x+\beta) = x^2 + (\alpha+\beta)x + \alpha\beta$

$\therefore x^2 + 2x + 6 = 0$ as $\alpha+\beta = -2$

$\alpha\beta = 6$

c) $(x^2-2)^2 - 7(x^2-2) = 0$

Let $w = x^2-2$

$\therefore w^2 - 7w = 0$

$\therefore w(w-7) = 0$

$\therefore w = 0, 7$

$\therefore x^2-2 = 0, 7$

$\therefore x^2 = 2, 9$

$\therefore x = \pm\sqrt{2}, \pm3$

Q7 a) (i) In $\triangle PST$ and $\triangle RUT$,

$\angle PTS = \angle UTR$ (vertically opposite $\angle$ s)

$\angle PST = \angle URT$ (alternate $\angle$ s on parallel lines)

$PS = RU$ (data)

$SP = UR$ (opposite sides of parallelogram)

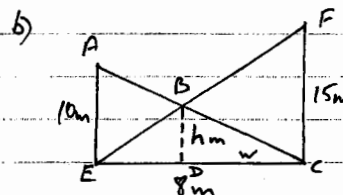
$\therefore RU = SP$

$\therefore \triangle PST \equiv \triangle RUT$ (AAS) QED

(ii) $ST = RT$ (corresponding sides in congruent $\triangle$ s)

$\therefore T$ is the midpoint of SR

QED



Let $DC = w$

(i) Now $\frac{w}{h} = \frac{8}{10}$

$\therefore w = \frac{4h}{5}$

(ii) $\frac{8-w}{h} = \frac{8}{15}$

$\therefore \frac{8-\frac{4h}{5}}{h} = \frac{8}{15}$

$\therefore 8h = 15(8-\frac{4h}{5})$

$= 120 - 12h$

$\therefore 20h = 120$

$\therefore h = 6m$

It is interesting that if we change the distance between the poles, the value of h does not change. Try replacing the 8 with a variable (say d) to prove this.